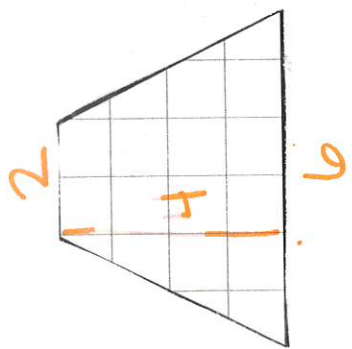


Add and invert a second trapezoid.

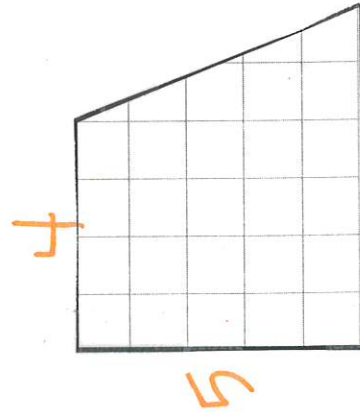
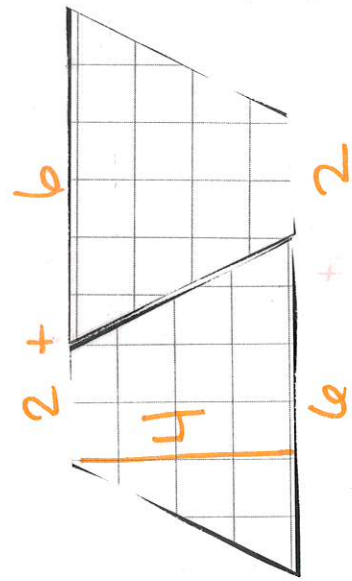
Decomposed/Rearranged Shape

Describe How to Find the Area

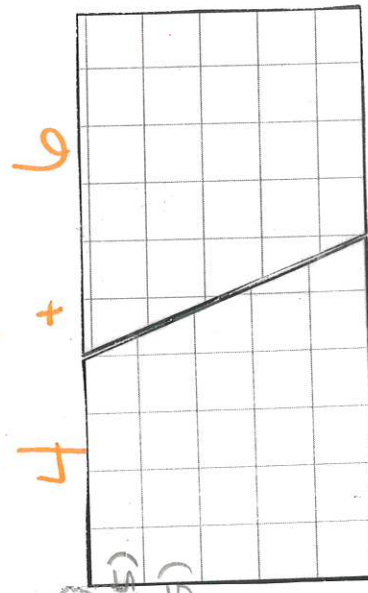
Original Shape



$$\begin{aligned}
 A &= bh \\
 A &= (2+6)(4) \\
 A &= \frac{1}{2}(2+6)(4) \\
 A &= \frac{1}{2}(8)(4) \\
 A &= 4(4) + \\
 A &= 16u^2
 \end{aligned}$$



$$\begin{aligned}
 A &= bh \\
 A &= (4+10)(5) \\
 A &= \frac{1}{2}(4+10)(5) \\
 A &= \frac{1}{2}(14)(5) \\
 A &= 5(5) \\
 A &= 25u^2
 \end{aligned}$$



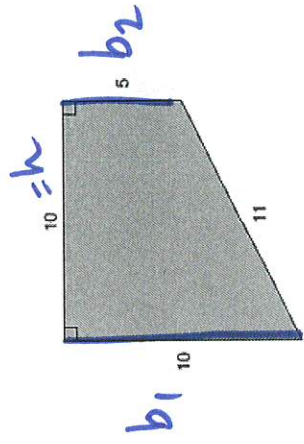
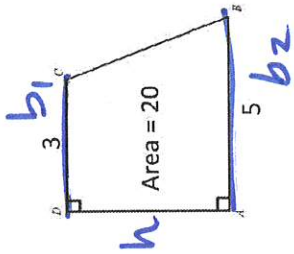
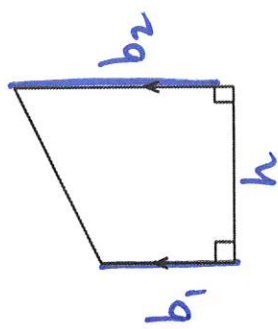
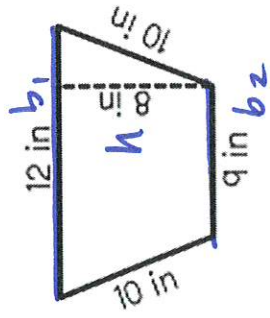
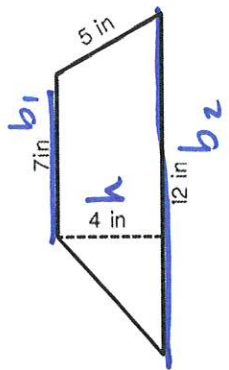
Add the two bases together and multiply by the height. Then multiply by half.

This is the same as dividing by 2.

Formula for Area of a Trapezoid:

$$A = \frac{1}{2}(b_1 + b_2)(h) \text{ or } A = \frac{(b_1 + b_2)(h)}{2}$$

The two bases of a trapezoid are ALWAYS the two _____ sides. Highlight the bases of each trapezoid. Use a different color to highlight the height.

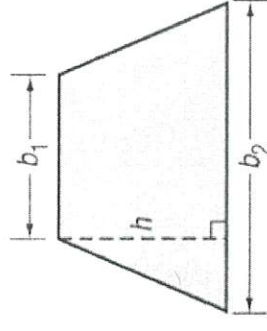


A trapezoid has two bases, b_1 and b_2 .

The height of a trapezoid is the distance between the two bases.

The area, A , of a trapezoid equals half the product of the height, h ,

and the sum of the bases b_1 and b_2 .



Base 1 14 Base 2 7

Height 5

Formula for area of a trapezoid:

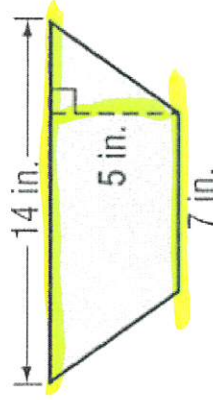
$$\frac{1}{2} (b_1 + b_2) h$$

$$\frac{1}{2} (14 + 7)(5)$$

$$\frac{1}{2} (21)(5)$$

$$\frac{1}{2} (105) = 52.5 \text{ in}^2$$

Substitute Values:



Base 1 7 Base 2 26

Height 12

Formula for area of a trapezoid:

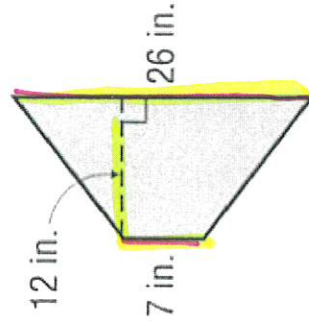
$$A = \frac{1}{2} (b_1 + b_2) h$$

$$A = \frac{1}{2} (7 + 26)(12)$$

$$A = \frac{1}{2} (33)(12)$$

$$A = 198 \text{ in}^2$$

Substitute Values:



Base 1 8 Base 2 18

Height 13.5

Formula for area of a trapezoid:

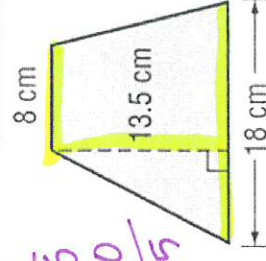
$$A = \frac{1}{2} (b_1 + b_2) h$$

$$A = \frac{1}{2} (8 + 18)(13.5)$$

$$A = \frac{1}{2} (26)(13.5)$$

$$A = 175.5 \text{ cm}^2$$

Substitute Values:



Base 1 0.4 Base 2 0.9

Height 0.8

Formula for area of a trapezoid:

$$A = \frac{1}{2} (b_1 + b_2) h$$

$$A = \frac{1}{2} (0.4 + 0.9)(0.8)$$

$$A = \frac{1}{2} (1.3)(0.8)$$

$$A = 0.52 \text{ m}^2$$

Substitute Values:

